| | | |
|---|---|---|
| **5(a)(i)** | work done per unit charge where electrical energy is converted into other forms of energy as the charge passes from one point to another | B1 |
| **5(a)(ii)1.** | $V = -1.5I + 6$ | B1 |
| **5(a)(ii)2.** | $E$ = 6.0 V<br>$r$ = 1.5 Ω | A1<br>A1 |
| **5(a)(iii)** | $R = \frac{V}{I} = \frac{5.4}{0.4} = 13.5\ \Omega$<br>$P = Vi = (5.4)(0.4) = 2.16$ W | A1<br>A1 |
| **5(b)** | $I = nAvq$<br>$\frac{I}{nq} = \text{constant} = Av$<br>$A_1 v_1 = A_2 v_2$<br>$\frac{v_{\text{rod}}}{v_{\text{case}}} = \frac{A_{\text{case}}}{A_{\text{rod}}} = \frac{h(\text{circumference of case})}{h(\text{circumference of rod})} = \frac{\pi d_{\text{case}}}{\pi d_{\text{rod}}}$<br>$= \frac{15}{1.5} = 10$ | M1<br>A1 |
| **6(a)** | electric field perpendicular to magnetic field | B1 |